# MarginMix

## Decision Engine — Scoring & Rules Logic

Complete reference for the deterministic margin risk assessment system

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This reference documents the standalone full assessment's deterministic Model1 computation. The scorecard is authoritative for question points, signal groups, arithmetic means, weights and verdict thresholds. The 51.7 example is a regression fixture, not a target for other answers. AI does not calculate or supply scores.

## LAYER 1 — QUESTIONS TO SIGNALS

The scorecard assigns points to 18 answers. Q1–Q6 are identifiers/context and are not in the composite. Q23 is a confidence override, not a numeric dimension input. Q27 is narrative-only. The full assessment is independent of the homepage quick profiler; answers are not prefilled or skipped.

### Context Questions — Not in the Composite

*Q1 Full Name · Q2 Work Email · Q3 Role / Title · Q4 Organisation Name · Q5 Organisation Size · Q6 Industry · Q8 Context · conditional Q10b Delivery Model*

### Model1 Scored Answers

| Q# | Answer | Points |
|---|---|---|
| Q7 | Decision: new / renewal / exploratory; expansion / escalation; strategic exception | 0 / 100 / 50 |
| Q9 | Stage: new / under 6 months / renewal-expansion; 6–12 months / 12+ months | 50 / 0 |
| Q10 | Retainer; hybrid / T&M / managed; fixed fee / commission / outcome-based | 0 / 50 / 100 |
| Q11, Q12, Q14, Q16, Q17, Q18 | Low / medium / high | 0 / 50 / 100 |
| Q13 | Minimal / periodic / frequent / continuous senior involvement | 0 / 50 / 100 / 100 |
| Q15 | Execution / balanced / thinking; advisory or strategy; technical implementation | 0 / 50 / 100; 100; 50 |
| Q19 | Junior execution / mid-level production / senior thinking / no clear path | 0 / 50 / 100 / 100 |
| Q20 | Significant / some / minimal / no meaningful additional value | 0 / 50 / 100 / 100 |
| Q21 | Less than usual / about the same / more than usual senior oversight | 0 / 50 / 100 |
| Q22 | Minimal / moderate / heavy coordination drag | 0 / 50 / 100 |
| Q24 | Agentic framework: no / yes / NA | 0 / 100 / 0 |
| Q25 | AI hours replaced: 0–25% / 25–50% / 50–75% / 75–100% / NA | 0 / 50 / 100 / 100 / 0 |
| Q26 | Commercial AI impact measured: yes / no / NA | 0 / 100 / 0 |

| | | |
|---|---|---|
| Q23 | Low confidence triggers the override; Q27 open text is narrative-only. | Override / no points |

### Scorecard interpretation

*Points are risk points, not dimension bands. Hybrid pricing is one answer worth 50 points; it is not separately decomposed into components. Fixed fee scores 100 across industries. Optional missing AI follow-ups use the existing NA mapping (0); unsupported answers fail explicitly rather than receiving invented points.*

## LAYER 2 — SIGNALS TO NUMERIC DIMENSIONS

Each named signal is first scored from its mapped answer. Signals joined by MAX use the highest point value. Each dimension is the arithmetic mean of its listed signal scores; no categorical dimension conversion is used for scoring.

| Dimension | Signal and source question | Aggregation |
|---|---|---|
| Workforce Intensity | SeniorDependency = MAX(Q13,Q21)<br>IterationLoad proxy = Q15<br>IterationLoad = Q16<br>GovernanceStrength = Q20 | Mean of 4 signal scores |
| Coordination Entropy | CoordinationIntensity = Q14<br>CoordinationLoad = MAX(Q18,Q22) | Mean of 2 signal scores |
| Commercial Exposure | EngagementState = MAX(Q7,Q9)<br>PricingRigidity = Q10<br>ScopeElasticity = Q17 | Mean of 3 signal scores |
| Volatility Control | ClientVolatility = Q11<br>StakeholderLoad = Q12 | Mean of 2 signal scores |
| Measurement Maturity | AILeverage = Q19<br>AIRiskPeak = MAX(Q24,Q25,Q26) | Mean of 2 signal scores |

## LAYER 3 — DIMENSIONS TO VERDICT

The raw composite is the sum of each numeric dimension mean multiplied by its unchanged weight. Preserve full precision through verdict selection; round only the displayed composite to one decimal place. The sample 51.7 is a fixture, not a score to impose on other answer sets.

### Step 1 — Hard Override (fires before scoring)

| Override | Condition | Verdict | Trigger Code |
|---|---|---|---|
| Confidence Failure | Fires when the existing Confidence Signal is Negative: low Q23 delivery confidence, or minimal/no additional value on Q20. This override is evaluated before the weighted verdict. | Do Not Proceed Without Repricing | LOW_CONFIDENCE_OVERRIDE |

### Step 2 — Weighted Score (applied when no override fires)

Dimension means are already numeric; do not convert them to bands and back into points. For each dimension, weighted contribution = dimension mean × weight ÷ 100. Composite = sum of the five contributions. The override does not replace or fabricate the underlying composite.

| Dimension | Weight | Score formula | Contribution |
|---|---|---|---|
| Workforce Intensity | 30% | Mean(Q13/Q21 MAX, Q15, Q16, Q20) | Mean × 0.30 |
| Coordination Entropy | 25% | Mean(Q14, Q18/Q22 MAX) | Mean × 0.25 |
| Commercial Exposure | 20% | Mean(Q7/Q9 MAX, Q10, Q17) | Mean × 0.20 |

| | | | |
|---|---|---|---|
| Volatility Control | 15% | Mean(Q11, Q12) | Mean × 0.15 |
| Measurement Maturity | 10% | Mean(Q19, Q24/Q25/Q26 MAX) | Mean × 0.10 |
| TOTAL | 100% | Sum of five contributions | Composite 0–100 |

### Illustrative regression fixture — not a target

| Dimension | Mean | Weight | Contribution |
|---|---|---|---|
| Workforce Intensity | 62.5 | 30% | 18.75 |
| Coordination Entropy | 50 | 25% | 12.5 |
| Commercial Exposure | 33.333… | 20% | 6.666… |
| Volatility Control | 25 | 15% | 3.75 |
| Measurement Maturity | 100 | 10% | 10 |
| Raw total | | | 51.666…; display 51.7 |

*Those values come from one fixed answer fixture. Other answers must produce their own signal scores, means and weighted contributions.*

### Score to Verdict Thresholds

| Score Range | Verdict | Trigger Code |
|---|---|---|
| 0 to <20 | Structurally Safe | WEIGHTED_SCORE_SAFE |
| 20 to <40 | Execution Heavy | WEIGHTED_SCORE_EXECUTION_HEAVY |
| 40 to <60 | Price Sensitive | WEIGHTED_SCORE_PRICE_SENSITIVE |
| 60 to <80 | Structurally Fragile | WEIGHTED_SCORE_FRAGILE |
| 80 to 100 | Do Not Proceed Without Repricing | WEIGHTED_SCORE_CRITICAL |

### Verdict Meanings

- **Structurally Safe:** Engagement viable under stated assumptions. Proceed with standard governance and monitoring.
- **Execution Heavy:** Senior dependency and iteration load will dominate effort. Elevated management attention required throughout delivery.
- **Price Sensitive:** High scope elasticity or pricing rigidity creates commercial vulnerability. Pricing assumptions require explicit protection.
- **Structurally Fragile:** High workforce intensity combined with high coordination entropy creates unsustainable margin pressure. Structural changes required.
- **Do Not Proceed Without Repricing:** Pricing or delivery confidence is insufficient to proceed safely. Engagement requires repricing or structural changes before commitment.

*The confidence override takes priority over every weighted-score band. There is no separate first-match structural-overload verdict rule.*

## CORE DESIGN PRINCIPLES

- **Deterministic scoring:** Every point, signal maximum, dimension mean and weighted contribution is calculated from the submitted answers. AI is limited to prose; numeric claims it produces are rejected.
- **Confidence override:** The existing negative Confidence Signal remains a pre-scoring override. It changes the verdict, not the numeric composite.
- **No margin forecast:** The assessment does not estimate margin erosion or effective margin, and does not invent staffing percentages or effort allocations.

• **Separate quick profiler:** The full assessment at /assessment has its own independent question, scoring and report flow. The homepage profiler does not prefill it or skip its questions.